

The Murray Reality Equation: A Falsifiable Phenomenological Law Relating Observer Information Processing to Quantum Measurement Statistics

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Abstract

The quantum measurement problem — explaining how definite outcomes emerge from quantum superpositions — remains unresolved after a century of debate. We propose the **Murray Reality Equation** (MRE), a minimal, falsifiable phenomenological law that quantitatively relates observer information processing to the statistical structure of binary quantum measurements. The MRE states that the alternation bias δ (negative lag-1 autocorrelation) in measurement outcomes depends on the observer’s information processing rate Φ according to:

$$\delta(\Phi) = \tanh[\operatorname{atanh}(\delta_0) + 2\alpha_0\kappa\Phi] \quad (1)$$

where δ_0 is the unobserved baseline bias, α_0 is a device-specific transfer factor (dimensions: s/bit), and κ is a proposed universal dimensionless coupling constant. We prove this formulation preserves relativistic no-signaling constraints, respects hard physical bounds, and is mathematically necessary if the observer influence is bounded and compositional. In the experimentally accessible small-signal regime ($2\alpha_0\kappa\Phi \ll 1$), this reduces to the testable linear approximation $\Delta\delta \approx 2\alpha_0\kappa\Phi$. We present a pre-registered experimental protocol using Geiger-Müller detectors and validated EEG entropy metrics, powered to detect $\kappa \geq 3 \times 10^{-6}$ or establish tight upper bounds. Three pre-specified falsification criteria ensure rigorous empirical testing of this paradigm-challenging hypothesis.

1 Introduction

1.1 The Persistent Measurement Problem

Quantum mechanics relies on two fundamentally incompatible evolution mechanisms. Isolated quantum systems evolve deterministically and unitarily according to the Schrödinger equation.

However, upon measurement, the quantum state undergoes a non-unitary, stochastic collapse governed by the Born rule. Standard quantum mechanics provides no rigorous definition of what constitutes a “measurement,” leaving this conceptual discontinuity — known as the Measurement Problem — unresolved for over a century [1, 2]. While decoherence theory successfully explains the transition from pure quantum superpositions to classical probability mixtures through environmental interactions [3], it still requires the Born rule to select a single definite outcome from that mixture. The question of *why* and *when* a particular outcome is realized remains fundamentally unanswered within the standard formalism. Numerous interpretations have been proposed — Many-Worlds [4], spontaneous collapse theories (GRW/CSL) [5, 6], QBism [7], consistent histories [8] — yet no consensus has emerged.

1.2 Observer-Dependent Approaches and Historical Context

Early quantum pioneers, including von Neumann [9], London and Bauer [10], and Wigner [11], proposed that consciousness might play a fundamental role in wavefunction collapse. However, these proposals remained qualitative and lacked testable predictions. The Princeton Engineering Anomalies Research (PEAR) laboratory conducted extensive experiments (1979–2007) on consciousness-related anomalies [12], but methodological limitations (no pre-registration, subjective intent, weak controls, poor reproducibility) prevented decisive conclusions [13].

1.3 The Need for Phenomenological Laws

Phenomenological laws — mathematical relationships describing observable correlations without presuming mechanisms — often precede mechanistic understanding (Planck’s law, Ohm’s law, Hubble’s law). A phenomenological approach is especially valuable for controversial hypotheses because it: (i) makes vague concepts measurable; (ii) gives falsifiable predictions; (iii) enables empirical progress despite incomplete theory; (iv) guides future theory via constraints. The MRE translates the qualitative “observer” notion into a quantitative framework with operational definitions and falsification criteria.

1.4 Core Innovation of the MRE

- (1) **Objective observer quantification:** Φ is the observer information processing rate (bits/s) from EEG CTW entropy, validated against behavioral information transfer.
- (2) **Rigorous device calibration:** α_0 is obtained from unattended automation, enabling cross-device extraction of a potentially universal κ .
- (3) **Bounded mathematical structure:** The tanh form enforces $|\delta(\Phi)| < 1$ and natural saturation.
- (4) **Mandatory falsification:** Three pre-specified criteria (F1–F3) provide decisive tests.

2 Mathematical Framework

2.1 The Core Functional Form

Definition 1 (Murray Reality Equation).

$$\delta(\Phi) = \tanh[\alpha \tanh(\delta_0) + 2\alpha_0 \kappa \Phi], \quad (2)$$

with $\delta \in (-1, 1)$, baseline δ_0 , device factor α_0 [s/bit], coupling κ [dimensionless], and observer rate Φ [bits/s].

2.2 Axiomatic Derivation

Theorem 2 (Uniqueness of Functional Form). *Equation (2) is the unique solution of: (i) $|\delta(\Phi)| < 1$ (boundedness); (ii) $\delta(\Phi_1 + \Phi_2) = \delta(\delta(\Phi_1), \Phi_2)$ (compositionality); (iii) smoothness in Φ .*

Proof. Let $g_\Phi : (-1, 1) \rightarrow (-1, 1)$ form a continuous one-parameter group. Then there exists a diffeomorphism $f : (-1, 1) \rightarrow \mathbb{R}$ and generator $\lambda = 2\alpha_0\kappa$ with $f(g_\Phi(x)) = f(x) + \lambda\Phi$, hence $g_\Phi(x) = f^{-1}(f(x) + \lambda\Phi)$. On $(-1, 1)$ the canonical chart is $f(x) = \text{atanh}(x)$ with inverse $\tanh$, yielding (2). $\square$

2.3 Differential Equation Form

$$\frac{d\delta}{d\Phi} = \lambda(1 - \delta^2), \quad \delta(0) = \delta_0, \quad \lambda = 2\alpha_0\kappa, \quad (3)$$

whose solution is (2). The factor $(1 - \delta^2)$ enforces asymptotic saturation.

2.4 Geometric Structure and Physical Analogues

2.4.1 Hyperbolic Geometry

In the Poincaré disk, $x \mapsto \tanh(\text{atanh } x + t)$ is a geodesic translation (isometry).

2.4.2 Special Relativity

Parallel velocity addition with $c = 1$ has identical composition: $\beta_{\text{tot}} = \tanh(\text{atanh } \beta_1 + \text{atanh } \beta_2)$.

2.4.3 Geometric Inevitability and Conventional Oversight

The functional form of Eq. (2) is *mathematically necessary* once two requirements are imposed: boundedness of the order parameter ($|\delta| < 1$) and linear composition of sequential influences (group action under addition of “effort” or Φ). These constraints force a conjugacy to addition on $\mathbb{R}$ via the Fisher- z chart $f(x) = \text{atanh } x$, yielding the $\tanh$ – atanh form uniquely. The same structure appears as geodesic translation in hyperbolic geometry and as relativistic velocity addition, suggesting a recurring organizational principle for bounded systems. Its relative absence from conventional quantum measurement modeling likely reflects a historical preference for linear or perturbative approximations, rather than the exact bounded group law that the geometry makes available. ¹

2.5 Small-Signal Linear Approximation

For $2\alpha_0\kappa\Phi \ll 1$ and $\delta_0 \approx 0$:

$$\Delta\delta \equiv \delta(\Phi) - \delta_0 \approx 2\alpha_0\kappa\Phi, \quad (4)$$

and in a transition GLM, the slope β_1 estimates $2\alpha_0\kappa$, giving

$$\hat{\kappa} = \frac{\beta_1}{2\alpha_0(1 - \delta_0^2)}. \quad (5)$$

¹See also community Q&A derivations emphasizing this mathematical inevitability [15].

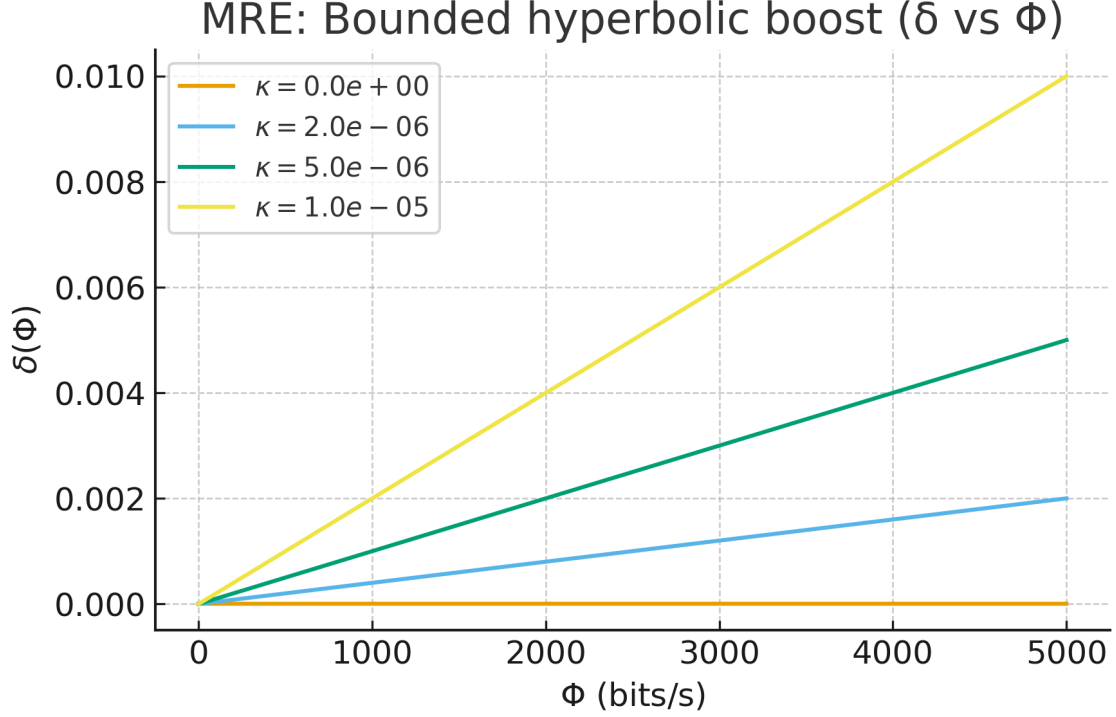


Figure 1: Bounded hyperbolic boost. The MRE prediction for $\delta(\Phi) = \tanh[\operatorname{atanh}(\delta_0) + 2\alpha_0\kappa\Phi]$ with $\delta_0 = 0$ and $\alpha_0 = 0.1$ s/bit across several κ . Curves are bounded ($|\delta| < 1$) and approach linearity for small $2\alpha_0\kappa\Phi$.

2.6 No-Signaling

Theorem 3 (Single Detector). *The stationary marginal $p = \Pr(X_t = 1)$ can be kept invariant under Φ variations while δ changes, preventing superluminal signaling.*

Proof. Let $q_0 = \Pr(1 | 0)$, $q_1 = \Pr(0 | 1)$; $\delta = q_0 + q_1 - 1$. Choosing (q_0, q_1) that vary with Φ while preserving the stationary p keeps p fixed as δ changes. Modeling in logit space ensures $q_x \in [0, 1]$ while decoupling δ from p . $\square$

Theorem 4 (Bipartite). *For spacelike-separated A, B with local CPTP dynamics and measurement independence, $\Pr(X_t^A)$ is independent of Φ^B (and vice versa).*

3 Operational Definitions

3.1 Alternation Bias

For binary $\{X_t\}_{t=1}^N$, $X_t \in \{0, 1\}$,

$$\delta = q_0 + q_1 - 1, \quad \hat{\delta} = \frac{\#(01) + \#(10) - \#(00) - \#(11)}{N - 1}. \quad (6)$$

Why δ ? It is the minimal statistic that changes the *temporal* structure (lag-1 autocorrelation) without touching the marginal rate p , which is required to preserve no-signaling.

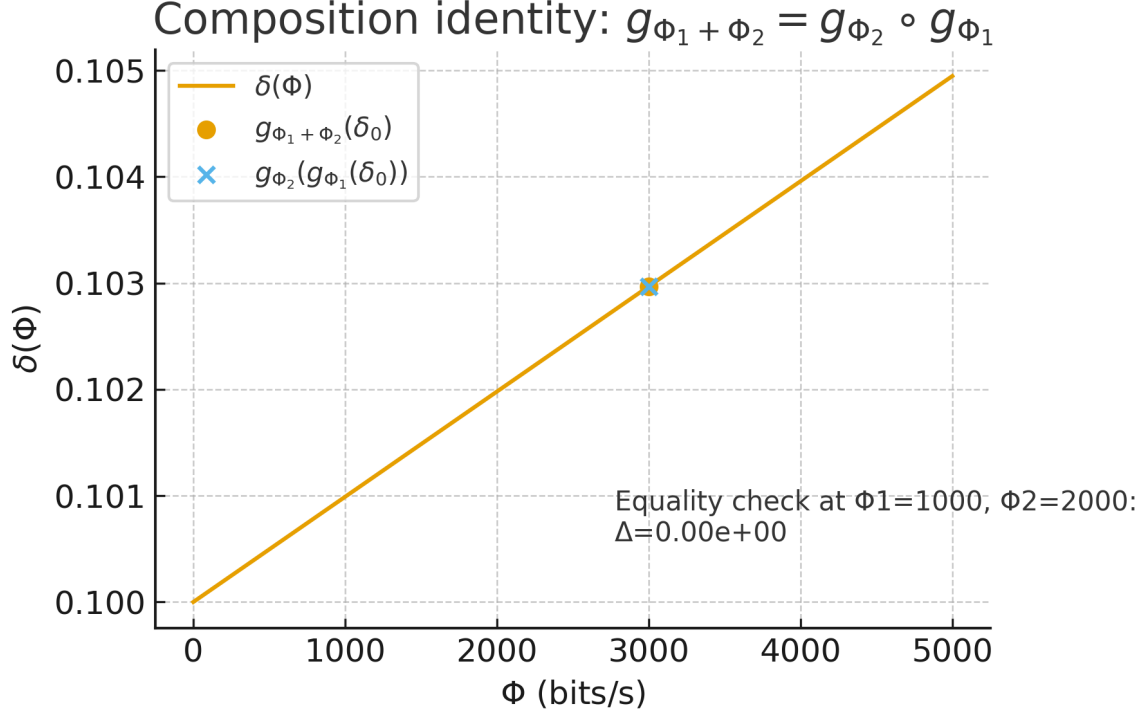


Figure 2: Composition identity. For $\delta_0 = 0.1$, $\alpha_0 = 0.1$ s/bit, $\kappa = 5 \times 10^{-6}$, we compare $g_{\Phi_1 + \Phi_2}(\delta_0)$ with $g_{\Phi_2}(g_{\Phi_1}(\delta_0))$ at $\Phi_1 = 1000$ bits/s, $\Phi_2 = 2000$ bits/s. Markers coincide, illustrating the one-parameter group action implied by Theorem 1.

3.2 Observer Information Rate Φ

OpenBCI Cyton (250 Hz), frontal montage (F3,Fz,F4,Fp1,Fp2). Preprocess: 1–40 Hz Butterworth (order 4); ICA artifact removal; average $\frac{1}{3}(F3 + Fz + F4)$. Entropy via CTW on 2 s windows (8-bit quantization). Define $\Phi(t) = H_{\text{CTW}}(t)/2$ s; smooth over 10 s. Validate with an oddball task requiring $\rho(\Phi, I(S; R)/T) > 0.7$. CTW is preferred over LZ or spectral entropy because it provides near-minimax, model-averaged entropy estimation for nonstationary, finite-memory processes and yields an immediate bits/s rate.

3.3 Device Transfer Factor α_0

Two-week unattended run at 10 Hz; median threshold; Laplace-smoothed conditional entropy $H(X_t | X_{t-1})$. Then $H_{\text{rate}}^{(\text{auto})} = H/\Delta t$ and $\alpha_0 = 1/H_{\text{rate}}^{(\text{auto})}$. The long automation averages out uncontrolled influences to establish a true baseline.

3.4 Coupling κ

Dimensionless; expected 10^{-6} – 10^{-5} . Extract via (5); nulls give upper bounds.

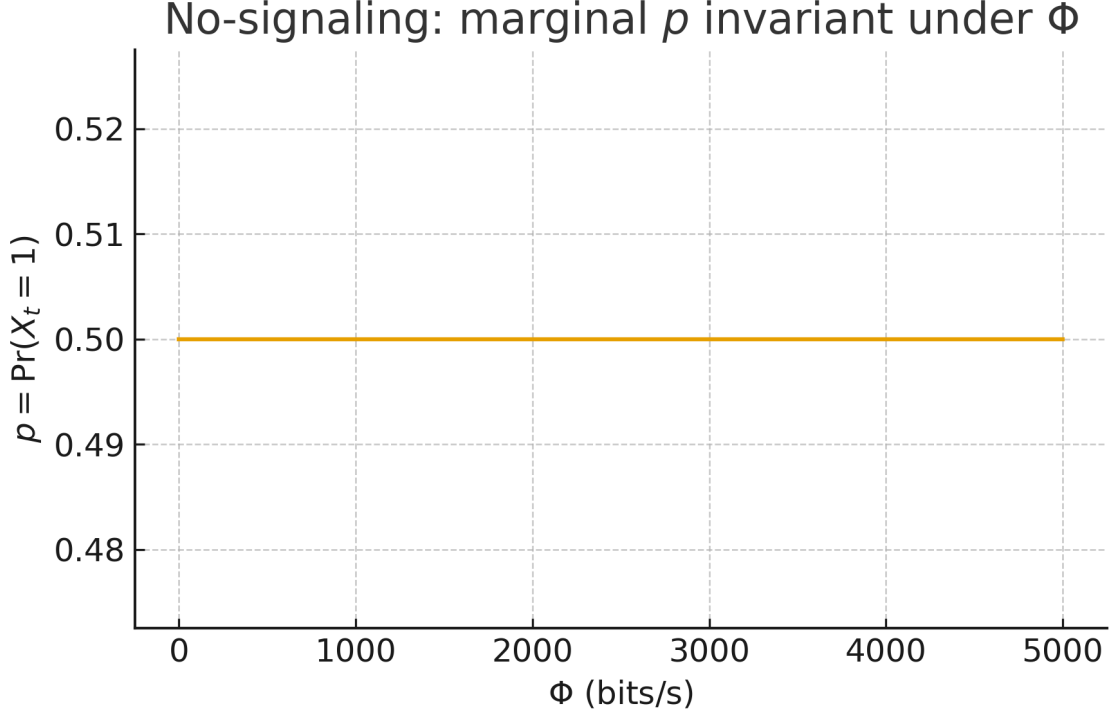


Figure 3: No-signaling I. With p defined to remain stationary (target $p = 0.5$), the marginal $p = \Pr(X_t = 1)$ is invariant as Φ varies (example: $\kappa = 5 \times 10^{-6}$, $\alpha_0 = 0.1$ s/bit).

4 Experimental Protocol

4.1 Hardware

Dual SBM-20 GM tubes with Am-241 sources ($\sim 1 \mu\text{Ci}$), mu-metal shielding, Teensy 4.1 (1 MHz timestamps), SD logging. EEG as above; environment: temperature (DS18B20), spare-EEG EMI, 3-axis accelerometer.

4.2 Binning

100 ms bins. $X_t = 1$ if count $\geq \theta$ (automation median), else 0. Robustness with $\theta \in \{40, 50, 60\}$ th percentiles. Control drift via block $\bar{p}_{\text{block}}$ (10 min).

4.3 Three Conditions

C1 (Automation). Two weeks unattended; estimate δ_0 , α_0 ; verify F1.

C2 (Focused attention). 60×30 min (30 h) with EEG; expect $\Phi \sim 10^3$ – 5×10^3 bits/s; test $\beta_1 > 0$.

C3 (Shuffled- Φ). Circularly shift $\Phi(t)$ by random lag > 60 s; expect $\beta_1 \approx 0$ (F2).

4.4 Power

Automation: $\sim 2 \times 10^7$ pairs; Focus: $\sim 2.2 \times 10^6$ pairs. With AR(1) $\rho \approx 0.6$, $N_{\text{eff}} \sim 2 \times 10^5$. Example parameters ($\kappa = 5 \times 10^{-6}$, $\alpha_0 = 0.1$ s/bit, $\Phi = 3000$ bits/s) give $\Delta\delta \approx 3 \times 10^{-3}$.

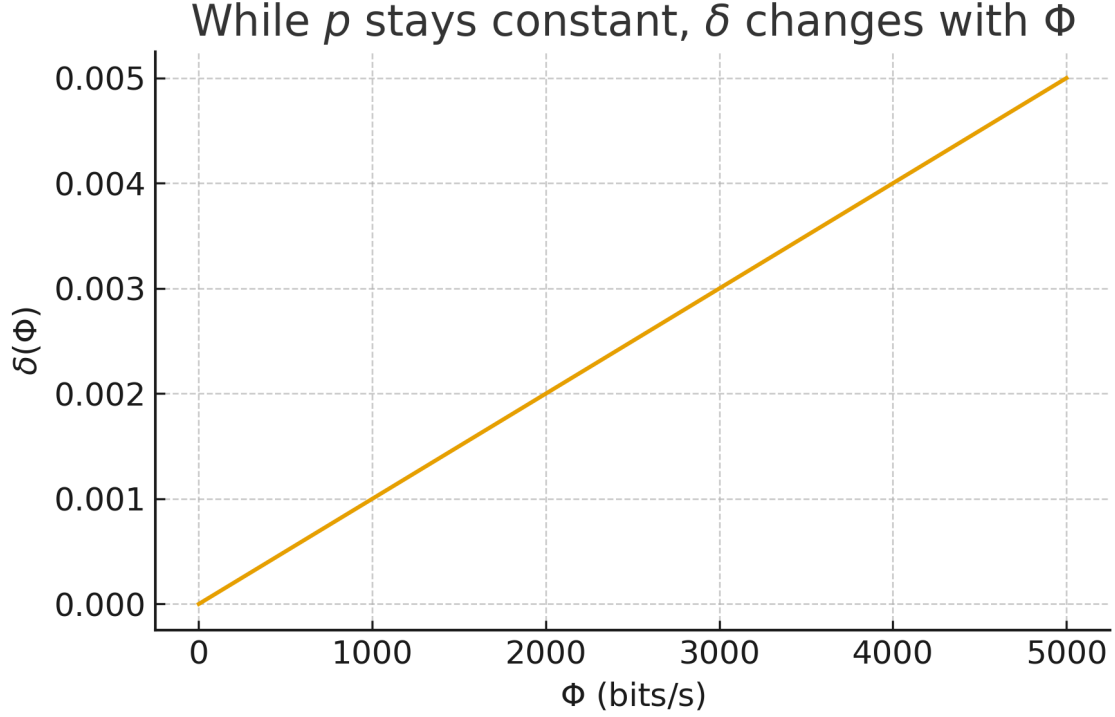


Figure 4: No-signaling II. Over the same Φ range, $\delta(\Phi)$ changes per the MRE. Together with Fig. 3a this shows temporal structure changes while the marginal stays fixed.

4.5 Analysis Plan

EEG pipeline as above; Φ from CTW; bitstream at 10 Hz; covariates: T_t , $\bar{p}_{\text{block}}$, circadian $\sin / \cos$.

Transition GLM:

$$\begin{aligned} \text{logit} \left[\Pr(X_t = 1 \mid X_{t-1} = x) \right] = & \beta_0^{(x)} + \beta_1^{(x)} \Phi_t \\ & + \gamma_1 T_t + \gamma_2 \bar{p}_{\text{block}} \\ & + \gamma_3 \sin(2\pi h/24) + \gamma_4 \cos(2\pi h/24). \end{aligned} \quad (7)$$

Inference by MLE, block bootstrap (block $L = 10\tau_{\text{auto}}$), and permutation (circular shifts). Claim requires perm- $p < 0.01$ and 99% bootstrap CI excluding zero.

Session-level: regress $\hat{\delta}_{\text{session}}$ on $\bar{\Phi}_{\text{session}}$ with HAC SEs.

Extract κ : use (5) and propagate SEs.

4.6 Falsifiers

F1: C1 slope $|\beta_1| < 2\text{SE}$. **F2:** Shuffled- Φ perm- $p > 0.10$. **F3:** Positive linear $\hat{\delta}$ vs. $\bar{\Phi}$ consistent with (4). Any single failure falsifies the MRE.

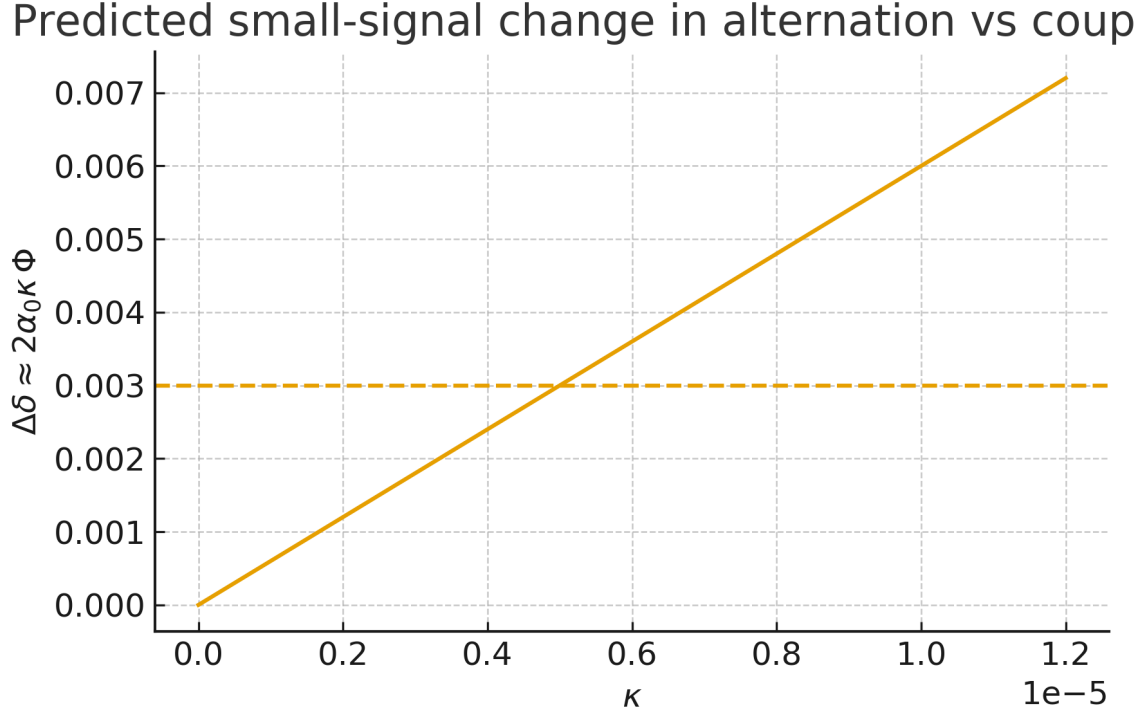


Figure 5: Small-signal scale. Predicted $\Delta\delta \approx 2\alpha_0\kappa\Phi$ for $\alpha_0 = 0.1$ s/bit and $\Phi = 3000$ bits/s vs κ .

5 Theoretical Implications and Connections

5.1 Quantum Interpretations

$\kappa \neq 0$ contradicts strictly unitary MWI (unchanged statistics), is potentially compatible with GRW/CSL (post-collapse statistics), quantifies the observer in Copenhagen, and offers an intersubjectively testable consequence in QBism.

5.2 Consciousness Theories

IIT proxy via Φ ; GWT alignment with global broadcasting; Orch-OR receives empirical impetus if any non-zero κ is confirmed (no microtubule assumption required here).

5.3 Phenomenology vs. Mechanism

The MRE is phenomenological; future work: derive mechanism/Hamiltonian, relate κ to constants, reconcile with biological decoherence, and construct QFT-consistent models.

Broader implications (non-claim, for future work)

If a non-zero κ is established, the compositional, bounded boost law in (2) would quantify a primitive linkage between cognitive information flow and physical outcome statistics. This invites — without presupposing — exploration of collective observation networks and possible roles for observer ensembles in cosmological evolution.

6 Discussion

6.1 Historical Context and Paradigm Challenge

Paradigm-challenging claims face heavy skepticism; testability and reproducibility decide outcomes. The MRE is framed to enable decisive empirical resolution.

6.2 Methodological Advances Beyond PEAR

Feature	PEAR	MRE
Pre-registration	No	Yes
Observer quantification	Subjective intent	Objective (EEG entropy)
Validation	None	Behavioral mutual information
Falsification criteria	None	Three mandatory (F1–F3)
Controls	Limited	Shuffled- Φ , automation
Reproducibility	Poor	Open data/code, kits for replication

Table 1: Methodological contrast.

6.3 Value Regardless of Outcome

Positive ($\kappa \neq 0$): new constant, possible framework revision, applications. Tight null: sharp constraints, methodological legacy, indirect support for unitary interpretations.

6.4 Limitations and Open Questions

No mechanism yet; single-subject initial study; device universality untested; decoherence puzzle remains. Future phases: replication, device diversity, distance/knowledge dependence, system dependence, theoretical derivation.

7 Research Roadmap

7.1 Phase 1 (Year 1): Initial Validation

Assemble, calibrate α_0 , run C1–C2, preregistered analysis, publish either way.

7.2 Phase 2 (Years 2–3): Replication/Extension

Independent replication, SPAD vs. GM for universality, extended controls, multi-observer additivity tests.

7.3 Phase 3 (Years 3–5): Mechanism

Probe decoherence/distance/knowledge dependence; other quantum platforms; theory building.

7.4 Phase 4 (5+): Applications

Conditional on confirmation: clinical tools, BCIs, RNGs, novel architectures.

8 Conclusion

The MRE quantifies a candidate observer influence on quantum measurement with four strengths: (i) rigorous bounded/compositional mathematics and no-signaling; (ii) operational clarity; (iii) decisive falsifiability; (iv) feasible experimentation. Either outcome — $\kappa \neq 0$ or tight null — yields substantive scientific value. The data decide.

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